

Exp - 03

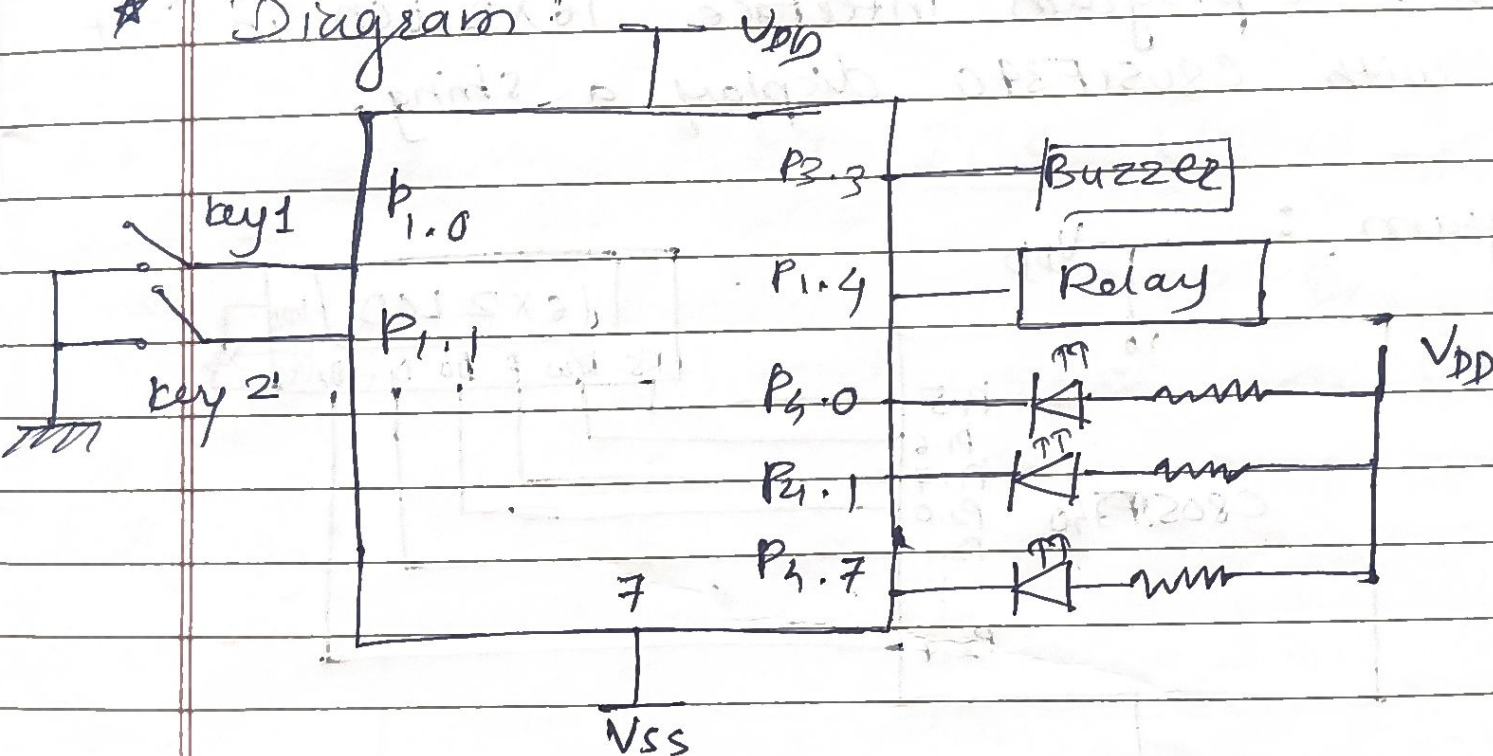
8051F340 has 5 I/O ports (P0-P4)

LED → active LOW Buzzer, Relay → Active HIGH

Problem Statement : →

C program to interface LEDs, buzzer & relay
Each o/p turns on when key is pressed.

Diagram :



Hardware Connections

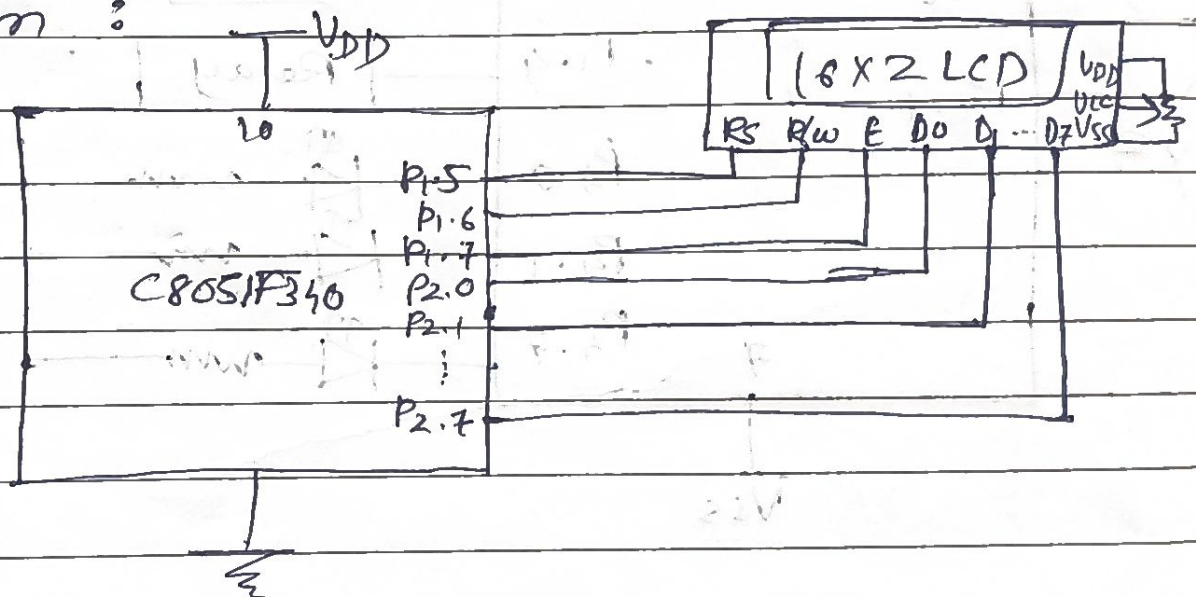
MCU Pin	ASK-25 Connector Pin	Signal	Active level
P1.0	PL3 - 1 (SW1)	Switch 1	low (pressed)
P1.1	PL3 - 2 (SW2)	Switch 2	low
P1.2	PL3 - 3 (SW3)	Switch 3	low
P1.3	PL3 - 4 (SW4)	Switch 4	low
P3.3	PL3 - 4 (Buzzer)	Buzzer	High (sound)
P1.4	PL3 - 7 (Relay)	Relay	Low (energized)
P4.0 - 7	PL6 - 10 ... 17 (LEDs 1-8)	LEDs	Low (on)
		3.3V	Power
		GND	Ground.

SFR	Description
P1MDIN	Port - 1 I/p mode : 0 → analog, 1 → digital
P1MDOUT	Port - 1 O/p mode : 0 → open drain, 1 → push
P3MDOUT	Port - 3 O/p mode (buzzer)
P4MDOUT	Port - 4 O/p mode (LEDs)
XBR2	Crossbar Enable : bit 6 = XBR2 = 1

• Exp - 04

★ Problem : C-program interface 16X2 LCD in 8-bit mode with 8051F340 display a string.

★ Diagram :



★ Hardware Connections

MCU Pin	LEP Pin	Signal
P1.5	RS	Register Select
P1.6	RW	Read / Write
P1.7	EN	Enable
P2.0	D0	Data bit 0
P2.7	D7	Data bit 7
GND	VSS	Ground
+5V	VCC	Power

★ SFR

P1MDOUT

P2MDOUT

XBR2

Configure P1.5-7 as push-pull
configure P2.0-7 as push pull
Crossbar Enable.

★ Formula :-

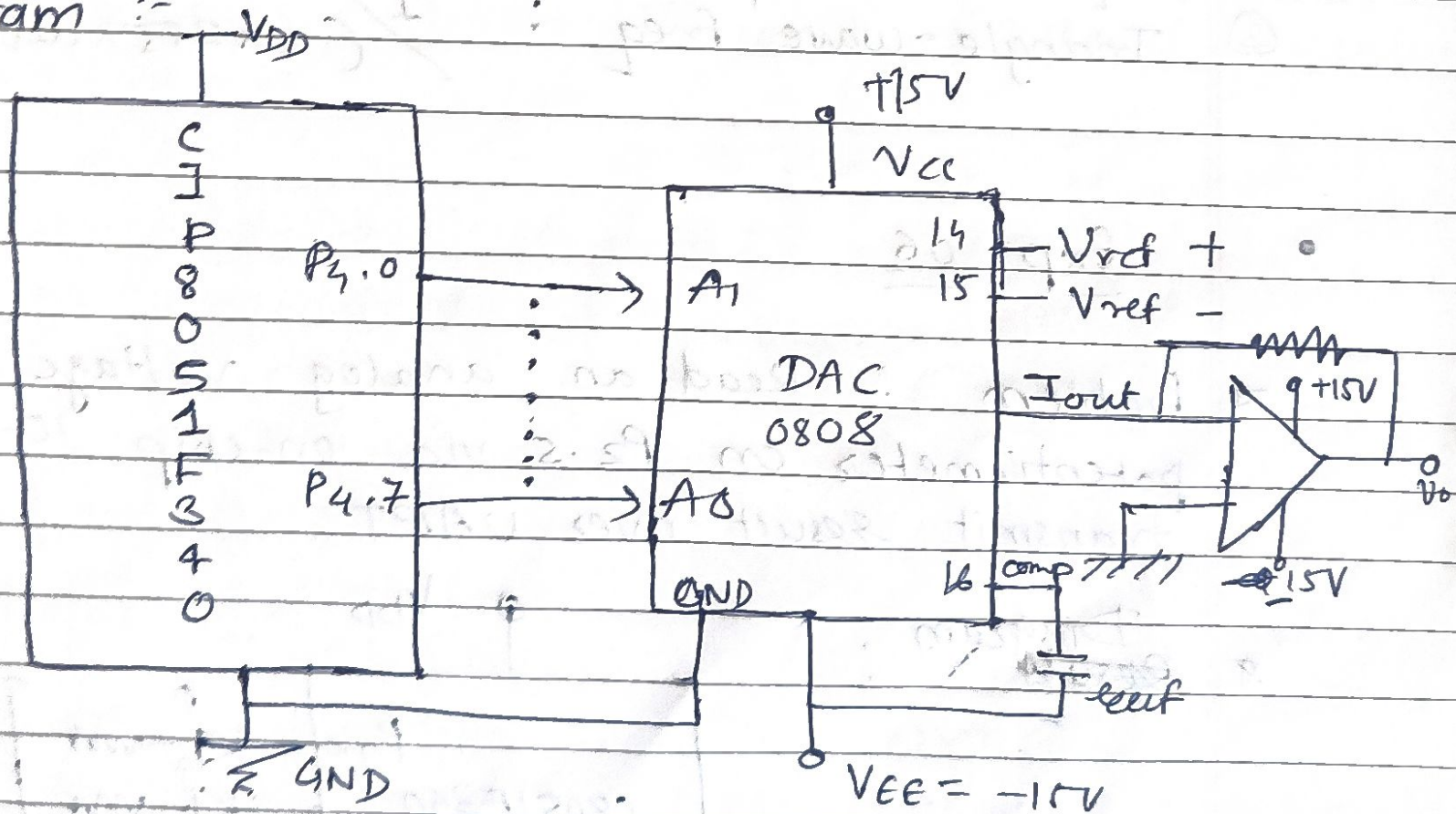
- ① LCD data sheet timings (clear = 2ms, commands = 40-50μs)
- ② Busy-flag polling can replace delay if R/W/read.

Exp 05

• Interfacing 8-bit DAC (Wave generation)

★ Problem :- Interface an R-2R DAC with 8051F340 & generate square, triangular waveforms

★ Diagram :-



★ Hardware Connections

MCU Pin	DAC Pin	Signal
P4.0	Do	Data bit 0
⋮	⋮	⋮
P4.7	D7	Data bit 7
±15V	V _±	Power
GND	GND	Ground

★ Relevant SFR

SFR	Description
P4MOUT	P4.0-7 push-pull
XBR2	Crossbar Enable

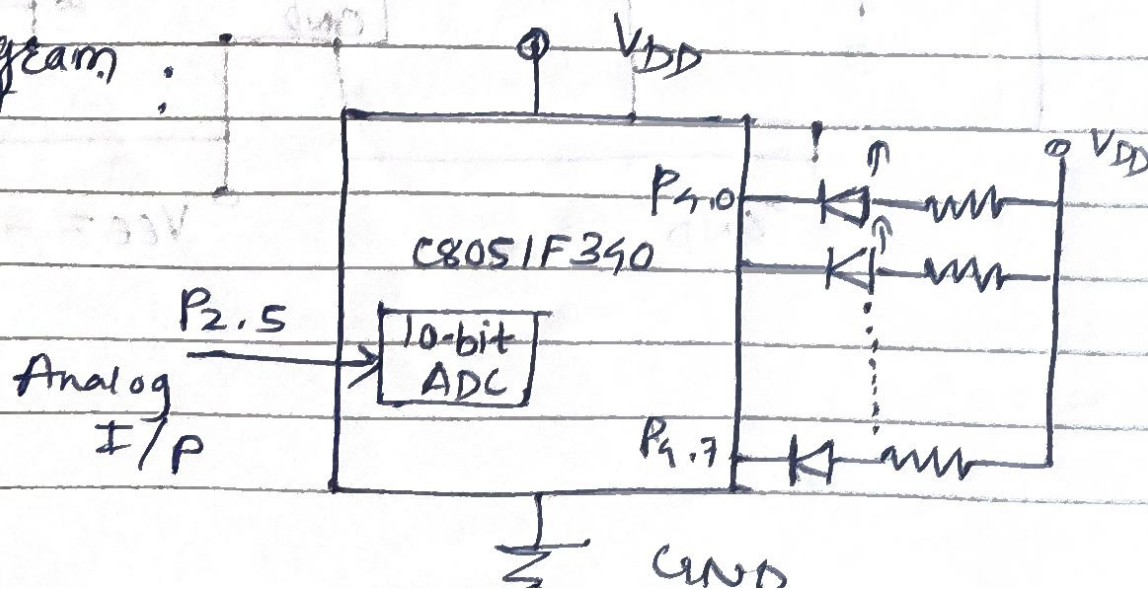
★ Formula

- ① DAC step size : $V_{ref}/256$
- ② Square wave period : $T = 2 \times \text{delay}$
- ③ Triangle-wave freq : $1/(2 \times 255 \times \text{delay-ms})$

Exp 06

★ Problem : Read an analog voltage from a potentiometer on P2.5 via on-chip 10-bit ADC & transmit result over UART.

★ Diagram :



★ Hardware Connections

MCU Pin	Signal	Destination
P2.5	AIN7	Pot wiper
GND		
P0.4	TXD	PC RX
P0.5	RXD	PC TX

★ SFR

REFOCN

Reference control

ADCOF

ADC clock & gain

ADCON

ADC enable & busy flag

AMXOP

ADC positive i/p select

XBRO/2

Enable UART pins

SCON

UART control

TMOD/TH1

Timer 1 for baud generation

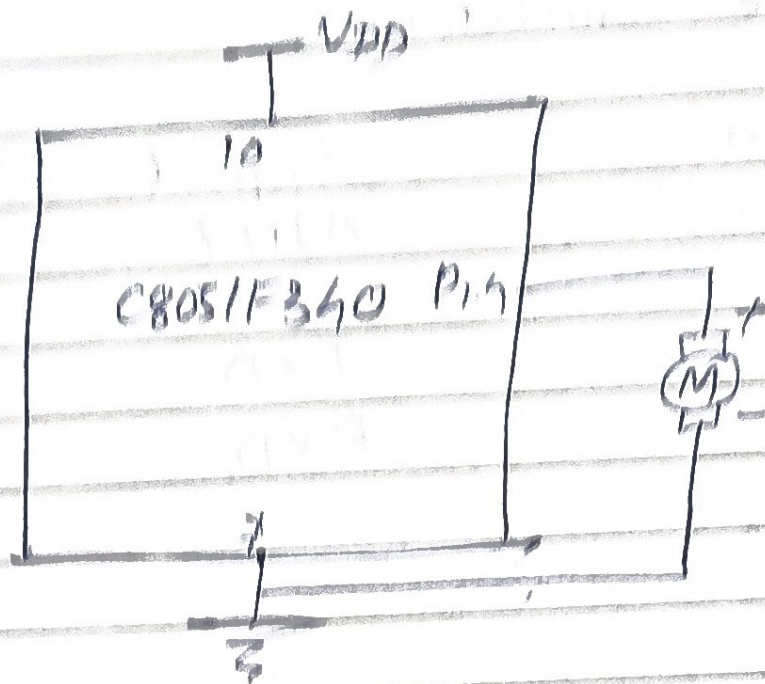
★ Formula :

① Step size $\rightarrow V_{ref} / 1024$ ② Digital result $\rightarrow \frac{V_{in}}{V_{ref}} \times 1023$

• Exp-07

★ Problem : To generate a PWM signal on PCA module (P1.4) of C8051F340 & control DC motor speed.

★ Diagram



★ Hardware Connections

MCU Pin	Function	Connection
PO.4	TxD	Bridge Rx → PC
PO.5	RxD	Bridge Tx ← PC
GND	Ground	Bridge Gnd

★ Relevant SFR

SCON0	Serial control
TMOD	Timer mode
TH1	Timer 1 load for baud
TR1	Timer 1 run control